

1) Appendix

1.1) Moore Penrose Pseudo Inverse (!exam)

Definition 1 (Moore Penrose Pseudo Inverse).

The pseudo inverse of a $N \times M$ matrix A is the unique matrix $A^\dagger$ satisfying:

$$\begin{aligned}AA^\dagger A &= A \\ A^\dagger A A^\dagger &= A^\dagger\end{aligned}$$

such that $AA^\dagger, A^\dagger A$ are both hermitian. $((AA^\dagger)^* = AA^\dagger, (A^\dagger A)^* = A^\dagger A)$

- **Tall Matrix** ($N > M$): If $A^* A$ is invertible then

$$A^\dagger = (A^* A)^{-1} A^*$$

is the left inverse of A ($A^\dagger A = I$)

- **Fat Matrix** ($N < M$): If AA^* is invertible then (right inverse)

$$A^\dagger = A^* (AA^*)^{-1}$$

is the right inverse of A ($AA^\dagger = I$)

- **Square Matrix**: If the matrix is square (and invertible) the pseudo inverse is the regular inverse.

Theorem 2 (Characterization of Least-Squares Solutions).

$\forall x \in \mathbb{K}^n$ we have $\|Ax - b\| \geq \|Az - b\|$ where $z = A^\dagger b$.

Moore Penrose Pseudo Inverse solves the least squares problem.

The disequality holds as an equality iff $x = A^\dagger b + (I - A^\dagger A)\vec{w}, \forall \vec{w}$ and provides infinite solutions unless A has full column rank (in this case $I - A^\dagger A = 0$). The solution with minimum euclidean norm is z .

Theorem 3 (Minimum Norm Solution).

If $Ax = b$ is satisfiable, the vector $z = A^\dagger b$ is a solution and satisfies $\|z\| \leq \|x\|$ for all solutions

2) Chapter 1: A Primer on Information Theory and MMSE

2.1) Information Content

Entropy:

$$\mathcal{H}(x) = - \sum_x p_x(x) \log_2 p_x(x) = -\mathbb{E}[\log_2 p_x(x)]$$

Properties:

- non negative
- maximized by $\sim u[a, b]$
- $\mathcal{H}(x) \leq \mathcal{H}(x|y)$
- $\mathcal{H}(x + c) = \mathcal{H}(x)$

Differential Entropy:

$$h(x) = - \int_x f_x(x) \log_2 f(x) dx = -\mathbb{E}[\log_2 f_x(x)]$$

Properties:

- Can be negative
- maximized by $\sim \mathcal{N}_{\mathbb{C}}(\mu, \sigma^2)$
- Cannot be approached by discretizing $f(x)$ ($h(x) = H(x^\Delta) + \log_2 \Delta$)

Proof $h(x)$ can be negative:

Let $x \sim u[0, b]$, then $h(x) = \int_0^b \frac{1}{b} \log_2 b \, dx = \log_2 b < 0 \, \forall b \in (0, 1)$

□

Mutual Information:

$$\begin{aligned} I(s; y) &= \mathcal{H}(s) - \mathcal{H}(s|y) \\ &= \mathcal{H}(y) - \mathcal{H}(y|x) \\ &= D(p_{sy} || p_s p_y) \end{aligned}$$

$$\begin{aligned} I(s; y) &= h(s) - h(s|y) \\ &= h(y) - h(y|x) \\ &= D(f_{sy} || f_s f_y) \end{aligned}$$

Mutual information can be obtained by discretizing x, y and calculating their entropy.

2.2) Reliable Communication

Channel Law (vectors)

$$f_{y|s}(y|s)$$

Then the mutual information becomes:

$$I(y; s) \stackrel{\text{memoryless}}{=} \mathbb{E} \left[\log_2 \frac{f_{y|s}(y|s)}{f_y(y)} \right] \stackrel{\text{memoryless+stationary}}{=} \mathbb{E} \left[\log_2 \frac{f_{y|s}(y|s)}{\frac{1}{2^{N_b}} \sum_{i=0}^{2^{N_b}-1} f_{y|s}(y|s_i)} \right]$$

Capacity:

Capacity is the maximum rate at which information can be transmitted over a channel with arbitrarily small error probability given an appropriate coding scheme. That is: $\exists$ coding scheme such that $p_e \rightarrow 0 \iff N \rightarrow \infty$. When this limit is reached, the error probability quickly grows.

$$C \stackrel{\text{ergodic, static, memoryless}}{=} \max I(s; y) \stackrel{\text{ergodic, static}}{=} \max \lim_{N \rightarrow \infty} \frac{1}{N} I(s; y)$$

Information Stable:

A channel is said to be information stable if the information density does not deviate asymptotically from the mutual information, that is: $\lim_{N \rightarrow \infty} \frac{1}{N} I(s; y) = \lim_{N \rightarrow \infty} \frac{1}{N} i(s; y)$

Where information density is the value whose expectation is the mutual information.

$$i(s; y) = \log_2 \frac{f_{y|s}(y|s)}{f_y(y)f_s(s)} \rightarrow I(s; y) = \mathbb{E}[i(s; y)]$$

Coding:

With finite length codewords the spectral efficiency $\left[\frac{\text{bit/s}}{\text{Hz}} \right]$ deviates from the capacity.

$$\frac{R}{B} - \sqrt{\frac{V}{N}} Q^{-1}(p_e) + O\left(\frac{\log N}{N}\right) \leq C$$

Where V is the variance of the information density.

2.3) MMSE Estimation, I-MMSE and LMMSE

The MMSE estimator chooses $\hat{s}(t)$ that Minimizes the Mean Square Error.

$$\hat{s}(y) = \mathbb{E}[s|y] \quad MMSE(\hat{s}(y)) = \mathbb{E}[|s(y) - \hat{s}(y)|^2] = \text{var}(\hat{s}(y))$$

Properties:

1. Unbiased in the mean: $\mathbb{E}[\hat{s}] = \mathbb{E}[\mathbb{E}[s|y]] = \mathbb{E}[s]$
2. Biased for specific implementation: $\mathbb{E}[\hat{s}|s] = \mathbb{E}[\mathbb{E}[s|y]|s] \neq s$
3. Satisfies Orthogonality Principle: $\mathbb{E}[g(y)(s(y) - \hat{s}(y))] = 0$, that is error is uncorrelated to function of observation
4. In AWGN it is a monotonically decreasing function of ρ (SNR)

Proof of 3:

Let the classic AWGN channel, then

$$\mathbb{E}[\hat{s}(y)|s] = \mathbb{E}\left[\frac{\sqrt{\rho}}{1+\rho}y|s\right] = \frac{\sqrt{\rho}}{1+\rho}\mathbb{E}[y|s] = \frac{\sqrt{\rho}}{1+\rho}s = s - \frac{1}{1+\rho} \neq s$$

□

Proof of 2:

$$\mathbb{E}[g(s - \hat{s})] = \mathbb{E}[\mathbb{E}[g(s - \hat{s})|y]] = \mathbb{E}[g\mathbb{E}[s|y] - g\mathbb{E}[\hat{s}|y]] \stackrel{\text{From 1.}}{=} \mathbb{E}[g\mathbb{E}[s] - g\mathbb{E}[s]] = 0$$

□

Proof that estimator is true given 3:

Compute MSE:

$$\begin{aligned} |s - \hat{s}|^2 &= |(s - \mathbb{E}[s|y]) + (\mathbb{E}[s|y] - \hat{s})|^2 \\ &= |s - \mathbb{E}|^2 + |\mathbb{E} - \hat{s}|^2 + 2\Re[(s - \mathbb{E})(\mathbb{E} - \hat{s})^*] \\ &= |s - \mathbb{E}|^2 + |\mathbb{E} - \hat{s}|^2 + 2\Re[g(y)(s - \mathbb{E})] \end{aligned}$$

where $g(y) = (\mathbb{E}[s|y] - \hat{s}(y))$. Now apply the expectation and the last term is the orthogonality principle:

$$\mathbb{E}[|s - \hat{s}|^2] = \mathbb{E}[|s - \mathbb{E}[s|y]|^2] + \mathbb{E}[|\mathbb{E}[s|y] - \hat{s}|^2] + 0$$

and is minimized when $\hat{s}(y) = \mathbb{E}[s|y]$.

□

With Know Channel:

$$f_{s|y} = \frac{f_{y|s}f_s}{f_y} = \frac{f_{y|s}f_s}{\int f_{y|s}f_s ds} \rightarrow \hat{s}(y) = \mathbb{E}[s|y] = \int s f_{s|y} ds = \frac{\int s f_{y|s}f_s ds}{\int f_{y|s}f_s ds}$$

I-MMSE

Relates MMSE of gaussian to I.

Scalar case:

$$MMSE(\rho) = \frac{1}{\log_2 e} \frac{d}{d\rho} I(s; y) = \frac{1}{\log_2 e} \frac{d}{d\rho} \log_2(1 + \rho) = \frac{1}{1 + \rho}$$

$$I(s; y) = \log_2 e \int_0^\rho MMSE(\rho) d\rho = \log_2 e \log(1 + \rho) = \log_2(1 + \rho)$$

Vector case:

$$\rho AE = \frac{1}{\log_2 e} \nabla_A I(s; y)$$

L-MMSE

Linear estimate for MMSE, In gaussian noise LMMSE=MMSE, otherwise good approximation to simplify model.

$$\hat{s}(y) = W_{MMSE}^* y + b$$

$$W_{MMSE} = R_y^{-1} R_{ys}$$

$$E = R_s - R_{ys}^* R_y^{-1} R_{ys}$$

2.4) Gaussian And AWGN

Gaussian Distribution

Complex	Scalar	Vector
Distribution	$x \sim \mathcal{N}_{\mathbb{C}}(\mu, \sigma^2)$	$x \sim \mathcal{N}_{\mathbb{C}}(\mu, \sigma^2)$
Distribution Function	$f_x(x) = \frac{1}{\pi\sigma^2} \exp \left\{ -\frac{\ x - \mu\ ^2}{\sigma^2} \right\}$	$f_x(x) = \frac{1}{\pi\sigma^2} \exp \left\{ -\frac{\ x - \mu\ ^2}{\sigma^2} \right\}$
Differential Entropy	$h(x) = \log_2(\pi e \sigma^2)$	$h(x) = \log_2(\pi e \sigma^2)$

Complex	Scalar	Vector
Differential Entropy Proof	$ \begin{aligned} h(x) &= -E[\log_2 f_x(x)] \\ &= -\mathbb{E}[-\log_2 \pi \sigma^2 - a \log_2 e] \\ &= \log_2 \pi \sigma^2 + \log_2 e \mathbb{E}\left[\frac{\ x - \mu\ ^2}{\sigma^2}\right] \\ &= \log_2 \pi \sigma^2 + \log_2 e \cdot 1 \end{aligned} $	$ \begin{aligned} h(x) &= -E[\log_2 f_x(x)] \\ &= -\mathbb{E}[-\log_2 \det(\pi R) - a \log_2 e] \\ &= \log_2 \det(\pi R) + \log_2 e \mathbb{E}[(x - \mu)^* R^{-1} (x - \mu)] \\ &= \log_2 \det(\pi R) + \log_2 e \cdot \text{tr}(I) \\ &= \log_2 \det(\pi e R) \end{aligned} $ where $ \begin{aligned} \mathbb{E}[a^* R^{-1} a] &= \mathbb{E}[\text{tr}(R^{-1} a a^*)] \\ &= \text{tr}(R^{-1} \mathbb{E}[a a^*]) \\ &= \text{tr}(R^{-1} R) \\ &= \text{tr}(I) \end{aligned} $ and $ \begin{aligned} \log_2 e \cdot \text{tr}(I) &= \log_2 e^{\text{tr}(I)} \\ &= \log_2 \det(eI) \end{aligned} $
	□	□

AWGN

Complex	Scalar	Vector
Base Form	$y = \sqrt{\rho}x + z$	$y = \sqrt{\rho}As + z$
Distributions	$ \begin{aligned} x &\sim \mathcal{N}_{\mathbb{C}}(0, 1) \\ z &\sim \mathcal{N}_{\mathbb{C}}(0, 1) \\ y &\sim \mathcal{N}_{\mathbb{C}}(0, 1 + \rho) \\ y s &\sim \mathcal{N}_{\mathbb{C}}(\sqrt{\rho}s, 1) \\ s y &\sim \mathcal{N}_{\mathbb{C}}\left(\frac{\sqrt{\rho}}{1 + \rho}y, \frac{1}{1 + \rho}\right) \end{aligned} $	$ \begin{aligned} x &\sim \mathcal{N}_{\mathbb{C}}(0, R_s) \\ z &\sim \mathcal{N}_{\mathbb{C}}(0, R_z = I) \\ y &\sim \mathcal{N}_{\mathbb{C}}(0, \rho A R_s A^* + R_z) \\ y s &\sim \mathcal{N}_{\mathbb{C}}(\sqrt{\rho}As, R_z) \\ s y &\sim \mathcal{N}_{\mathbb{C}}(\sqrt{\rho}R_s A^* (\rho A R_s A^* + R_z)^{-1} y, (R_s^{-1} + \rho A^* R_z^{-1} A)^{-1}) \end{aligned} $
Mutual Information	$I(s; y) = \log_2(1 + \rho)$	$I(s; y) = \log_2 \det(I + \rho A R_s A^* R_z^{-1})$
Mutual Information Proof	$ \begin{aligned} I(y; s) &= h(y) - h(y s) \\ &= \log_2(\pi e(1 + \rho)) - \log_2(\pi e) \\ &= \log_2(1 + \rho) \end{aligned} $	$ \begin{aligned} I(y; s) &= h(y) - h(y s) \\ &= \log_2 \det(\pi e(\rho A R_s A^* + R_z)) - \log_2 \det(\pi e(R_z)) \\ &= \log_2 \det(I + \rho A R_s A^* R_z^{-1}) \end{aligned} $
	□	□
MMSE Estimator	$\hat{s}(y) = \frac{\sqrt{\rho}}{1 + \rho}y$	$\hat{s}(y) = \sqrt{\rho}R_s A^* (\rho A R_s A^* + R_z)^{-1}y$
MMSE	$MMSE(\rho) = \frac{1}{1 + \rho}$	$E = (R_s^{-1} + \rho A^* R_z^{-1} A)^{-1}$

Proof of MMSE Estimator and MMSE for scalar case:

Recall $\hat{s}(y) = \frac{\int s f_{y|s} f_s ds}{\int f_{y|s} f_s ds}$ and by noticing the exponents

$$f_{y|s} f_s = \frac{1}{\pi^2} e^{-|y - \sqrt{\rho} s|^2 - |s|^2} = \frac{K}{\pi \frac{1}{1+\rho}} \exp \left\{ -\frac{|s - \frac{\sqrt{\rho}}{1+\rho} y|^2}{\frac{1}{1+\rho}} \right\} \sim K \mathcal{N}_{\mathbb{C}} \left(\frac{\sqrt{\rho}}{1+\rho} y, \frac{1}{1+\rho} \right) \propto f_{\hat{s}|y}$$

And therefore

$$\hat{s}(y) = \frac{\mathbb{E}[f_s]}{1} = \frac{\sqrt{\rho}}{1+\rho} y$$

and the MMSE is the variance of $\hat{s}(y) \propto f_{\hat{s}|y}$

□

2.5) Additional Exercises

Example 1.2 (Entropy Of Bernoulli RV).

Express entropy of bernoulli rv:

$$x = \begin{cases} 0 & \text{with probability } p \\ 1 & \text{with probability } 1 - p \end{cases}$$

Solution:

Recall entropy: $H(x) = -\sum_x p(x) \log_2 p(x)$, then

$$H(x) = -p \log_2 p - (1 - p) \log_2 (1 - p)$$

Solve additional min/max problem:

$$\frac{d}{dp} H(x) = -\log_2(p) - \frac{1}{\ln p} + \log_2(1 - p) + \frac{1}{\ln(p)} = \log_2 \frac{1 - p}{p}$$

where $p = 0.5$ provides the max entropy $H_{\max} = 1$ and the min entropy is $p = 0 \implies H_{\min} = 0$ by changing the og entropy formula

Some derivatives:

$$\begin{aligned} \frac{d}{dx} \log_2 x &= \frac{1}{x} \frac{1}{\ln 2} \\ \frac{d}{dx} \log_k f(x) &= \frac{f'(x)}{f(x)} \frac{1}{\ln k} \\ \frac{d}{dx} f(x) \log_k f(x) &= f'(x) \left(\log_k f(x) + \frac{1}{\ln k} \right) \end{aligned}$$

Exercise (capacity of BSC).

The binary channel in which the input symbols are complemented with probability p is the simplest model of a channel with errors, yet it captures most of the complexity of the general problem. Calculate the capacity

Solution:

let $p_s(s) = \begin{cases} 0 & \text{with probability } p_s \\ 1 & \text{with probability } 1 - p_s \end{cases}$, then the channel law is: $p_{y|s}(y|s)$ and can be found in this way:

$$p_{y|s}(y|s) \begin{cases} p(0|s) = \begin{cases} p(0|0) = 1 - p \\ p(1|0) = p \end{cases} \\ p(1|s) = \begin{cases} p(1|0) = p \\ p(1|1) = 1 - p \end{cases} \end{cases}$$

and then the capacity is the maximum of the mutual information $I(s; y) = H(y) - H(y|s)$ where $H(y) = 1$ is maximized when $p_s = 0.5$. Now we must only calculate $H(y|s) = p_s(0)H(y|0) + p_s(1)H(y|1)$ by symmetry the two entropies are the same, so I just calculate one of them:

$$-H(y|0) = p(0|0)\log_2 p(0|0) + p(1|0)\log_2 p(1|0) = (1-p)\log_2(1-p) + p\log_2(p)$$

$$H(s; y) = 1 - 0.5[2(H(y|s))] = 1 + p\log_2 p + (1-p)\log_2(1-p)$$

And thus

$$C = \max I(s; y) \rightarrow 1$$

Extra

Example ($H(x)$ cannot be approached by discretizing f_x).

The entropy of a b -bit quantization is approximately $h(x) + b$, however this diverges as $b \rightarrow \infty$.

From the mean value thm we have that

$$\exists x_i \in [i\Delta, (i+1)\Delta] : f(x_i) \cdot \Delta = \int_{i\Delta}^{(i+1)\Delta} f(x)dx$$



Bad Graphical representation

Now consider $x^\Delta = x_i \in [i\Delta, (i+1)\Delta)$ by the mean value thm we have the discretized PMF $P[x_i] = \int f(x)dx = f(x_i)\Delta = p_i$. What happened is that by the mean val thm and by a small Δ we approximated $f \approx p$. This is exactly how integrals are computed, so this should work, right?

The entropy of this quantized rv is found by

$$\begin{aligned} H(x^\Delta) &= - \sum_i p_i \log_2 p_i \\ &= - \sum_i \Delta f(x_i) \log_2 (\Delta f(x_i)) \\ &= \sum_i \Delta f(x_i) (\log_2 \Delta + \log_2 f(x_i)) \\ &\text{notice how } \sum_i \Delta f(x_i) \log_2 \Delta = \log_2 \Delta \\ &= - \underbrace{\sum_i \Delta f(x_i) \log_2 f(x_i)}_{h(x)} - \log_2 \Delta \\ H(x^\Delta) &= h(x) - \log_2 \Delta \rightarrow h(x) = H(x^\Delta) + \log_2 \Delta \end{aligned}$$

Where $\sum \Delta f(x_i) = 1$ by definition of CDF.

This holds for a uniform rv $x \sim U(0,1)$ as $\Delta = 2^{-n}$ and thus $H(x^\Delta) = n$ but this doesn't hold for a gaussian rv as it needs more than n bits: $x \sim N(0, \sigma^2)$, $\Delta = 2^{-n}$ $H(x^\Delta) = \log(\pi e \sigma^2) + n$, so information is lost and the approximation is not valid.

□

Exercise (Discretization of X and Y can lend to mutual information calculation).

By recalling the result of [98221d](#) we can also find $H(X^\Delta|Y^\Delta)$.

In fact, by applying the mean val thm in 2D we have that

$$P_{x,y} = \iint f_{xy}(x,y) dx dy = \Delta_x \Delta_y f_{x,y} = p_{x,y}$$

And thus

$$p_{x|y} = \frac{p_{x,y}}{p_y} = \frac{\Delta_x \Delta_y f_{x,y}}{\Delta_y f_y} = \Delta_x \frac{f_{x,y}}{f_y} = \Delta_x f_{x|y}$$

And as before

$$\begin{aligned} H(X^\Delta|Y^\Delta) &= - \sum_x \sum_y p(x,y) \log_2 p(x|y) \\ &= - \sum_x \sum_y \Delta_x \Delta_y f(x,y) (\log_2 \Delta_x - \log_2 f(x|y)) \\ &= -\log_2 \Delta_x + \sum_x \sum_y \Delta_x \Delta_y (x,y) \log_2 f(x|y) \\ &= h(x|y) - \log_2 \Delta_x \end{aligned}$$

Then just compare:

$$I(x;y) = H(X^\Delta) - H(X^\Delta|Y^\Delta) = h(x) - \log_2 \Delta_x - (h(x|y) - \log_2 \Delta_x) = h(x) - h(x|y)$$

□

3) Chapter 2: A Signal Processing Perspective

3.1) Signal Representation

Baseband TXSignal:

$$x(t) = x_i(t) + jx_q(t)$$

Upconversion to Passband Signal:

$$\begin{aligned}x_p(t) &= \sqrt{2}\Re(x(t)e^{j2\pi f_c t}) \\&= \sqrt{2}\Re((x_i + jx_q)(\cos + j\sin)) \\&= \sqrt{2}(x_i(t)\cos(2\pi f_c t) - x_q(t)\sin(2\pi f_c t)) \\&= A(t)\cos(\phi(t))\cos(2\pi f_c t) + A(t)\sin(\phi(t))\sin(2\pi f_c t) \\&= A(t)\cos(2\pi f_c t + \phi(t))\end{aligned}$$

Passband RX Signal:

$$y_p(t) = \sqrt{2}(y_i(t)\cos(2\pi f_c t) - y_q(t)\sin(2\pi f_c t))$$

Downconversion to Baseband Signal:

Find $y_i(t)$ by multiplying and then low pass filter

$$y_p(t)\sqrt{2}\cos(2\pi f_c t) = y_i(t) + y_i(t)\cos(4\pi f_c t) - y_q(t)\sin(2\pi f_c t)$$

Same for $y_q(t)$

3.2) Channel Models, Normalization Pulse Shaping and Noise Properties

Channel Model

$$y_p(t) = (c_p * x_p)(t) = \int c_p(\tau)x_p(t - \tau)d\tau$$

c_p doesn't need to be band limited since x_p already is $\rightarrow$ also y_p is band limited

$\downarrow$ Downshift

$$c(\tau) = c_p(\tau)e^{-j2\pi f_c \tau} \rightarrow \text{used for one tap (frequency flat) channels}$$

$\downarrow$ Band Limit

$$g_{B/2} = B\text{sinc}(B\tau) \rightarrow c_b(\tau) = (g_{B/2} * c(\tau)) = B\text{sinc}(B\tau) * (c_p(\tau)e^{-j2\pi f_c \tau})$$

$\downarrow$ Discretize in Time

$$c[l] = Tc_b(\tau)|_{\tau=LT+\phi} \quad T \leq 1/B$$

Tapped Delay Line

$$\begin{aligned}
c_p(\tau) &= \sum_q A_q \delta(\tau - \tau_q) \\
c(\tau) &= \sum_q A_q \delta(\tau - \tau_q) e^{-j2\pi f_c \tau_q} \\
c_b(\tau) &= \sum_{q=0}^{Q-1} B A_q \text{sinc}(B(\tau - \tau_q)) e^{-j2\pi f_c \tau_q} \\
c[l] &= \sum_{q=0}^{Q-1} T B A_q \text{sinc}(B(lT + \phi - \tau_q)) e^{-j2\pi f_c \tau_q} \\
&\quad \text{Let } \phi = \tau_0 \\
&= A_0 \text{sinc}(l) e^{-j2\pi f_c \tau_0} + \sum_{q=1}^{Q-1} A_q \text{sinc}\left(l - \frac{\tau_0 - \tau_q}{T}\right) e^{-j2\pi f_c \tau_q}
\end{aligned}$$

and is frequency flat only iff $\tau_q = n\tau_0$, $n \in \mathbb{N}$.

Channel Normalization

SISO: $c[l] = \sqrt{G}h[l]$

$$\sum_{n=0}^{N-1} h[n] = 1$$

MIMO:

$$\begin{aligned}
\sum_{l=0}^L \sum_{i=0}^{N_r-1} \sum_{j=0}^{N_t-1} \mathbb{E}[|H[l]_{(i,j)}|] &= N_t N_r \\
\sum_{l=0}^L \mathbb{E}[|H[l]|_F] &= \sum_{l=0}^L \mathbb{E}[\text{tr}(H[l]H^*[l])] = N_r N_t \xrightarrow{\text{flat}} \sum_{l=0}^L \mathbb{E}[|H[l]|_F] = \mathbb{E}[\text{tr}(H[l]H^*[l])] = N_r N_t
\end{aligned}$$

Pulse Shaping:

$$x(t) = \sum_n x[n] \text{sinc}\left(\frac{t - nT}{T}\right)$$

Although sinc not really used. Raised cosine has penalty if $\frac{1}{1+b}$

Noise Properties:

- White since autocovariance is $R_v = \mathbb{E}[v[t]v^*[t - \tau]] = N_0\delta(t)$ and thus it is static and ergodic
- Gaussian since $v \sim \mathcal{N}_C(0, N_0 = kT_{eff})$ and by Law of Large Numbers sum of iid is gaussian
- Power Spectrum is $S_v(f) = N_0$ and thus in B it has power $P_v = \int_B N_0 = BN_0$

3.3) Vector Representations

SISO

$$y[n] = \sqrt{G} \sum_l h[l] x[n-l] + v[n] \xrightarrow{\text{Stacked Vector}} \mathbf{y}_N = \sqrt{G} \mathbf{H}_{N,N+L} \mathbf{x}_{N+L} + \mathbf{v}_N$$

where:

- $(\cdot)_N = [(\cdot)[0], \dots, (\cdot)[N-1]]^T$
- $\mathbf{x}_{N+L} = \underbrace{[x[-L], \dots, x[-1]]}_{\text{past values}}, x_N]^T$

- The Toeplitz matrix

$$\mathbf{H}_{N,N+L} = \begin{bmatrix} h[L] & \dots & h[0] & 0 & 0 & \dots & 0 \\ 0 & h[L] & \dots & h[0] & 0 & \dots & 0 \\ \vdots & & & \ddots & & & \vdots \\ 0 & \dots & \dots & 0 & h[L] & \dots & h[0] \end{bmatrix}$$

MIMO

i-th antenna with conv & TX-sum:

$$y^{(i)}[n] = \sqrt{G} \sum_{j=0}^{N_t-1} \sum_l h^{(i,j)}[l] x^{(j)}[n-l] + v^{(i)}[n]$$

MIMO base relation: all antennas with convolution:

$$y[n] = \sqrt{G} \sum_l H[l] x[n-l] + v[n]$$

where:

- $(\cdot)[n] = [(\cdot)^{(0)}[n], \dots, (\cdot)^{(N_r-1)}[n]]^T$
- $x[n] = [x^{(0)}[n], \dots, x^{(N_t-1)}[n]]^T$

- $H[l] = \begin{bmatrix} h^{(0,0)}[l] & \dots & h^{(0,N_t-1)}[l] \\ \vdots & \ddots & \vdots \\ h^{(N_r-1,0)}[l] & \dots & h^{(N_r-1,N_t-1)}[l] \end{bmatrix}$

i-th antenna with TX-sum:*

$$y_N^{(i)} = \sqrt{G} \sum_{j=0}^{N_t-1} H_{N,N+L}^{(i,j)}[l] x_{N+L}^{(j)}[n-l] + v_N^{(i)}[n]$$

using stacked vectors, that is:

- $(\cdot)_N^{(i)} = [(\cdot)^{(i)}[0], \dots, (\cdot)^{(i)}[N-1]]^T$
- $x_{N+L}^{(j)} = [\underbrace{x^{(j)}[-L], \dots, x^{(j)}[-1]}_{\text{past values}}, x_N^{(j)}]^T$

Final MIMO relation: all antennas and no convolution

$$y_{N_r N} = \sqrt{G} H_{N_r N, N_t(N+L)} x_{N_t(N+L)} + v_{N_r N}$$

where:

- $(\cdot)_{N_r N} = [(\cdot)_N^0, \dots, (\cdot)_N^{N_r-1}]^T$
- $x_{N+L} = [x_N^0, \dots, x_N^{N_t-1}]^T$

- $H_{N_r N, N_t(N+L)} = \begin{bmatrix} H_{N,N+L}^{(0,0)} & \dots & H_{N,N+L}^{(N_r,0)} \\ \vdots & \ddots & \vdots \\ H_{N,N+L}^{(0,N_t-1)} & \dots & H_{N,N+L}^{(N_r,N_t-1)} \end{bmatrix}$

3.4) Power And Precoding

Power & SNR Relations

$$P_r = G P_t = G E_s / T \quad SNR = \frac{P_r}{N_0 B} = \frac{G P_t}{N_0 / T} = \frac{G E_s}{N_0}$$

$$C(SNR) = \log_2(1 + SNR)$$

$$E_b = \frac{GP_t}{R} = \frac{GE_s}{R/B} = N_0 \frac{SNR}{R/B} = N_0 \frac{SNR}{C(SNR)} = N_0 \frac{SNR}{\log_2(1 + SNR)}$$

SISO Precoder

$$x[n] = \sqrt{E_s P[n]} s[n] \quad \frac{1}{N} \sum_n P[n] = 1$$

MIMO Precoder

$$x[n] = \sqrt{\frac{E_s}{N_t}} F[n] s[n] \quad N_s \leq N_t$$

moreover:

$$\begin{aligned} s[n] \text{ i.i.d.} &\rightarrow \mathbb{E}[s[n] s^*[n]] = I \\ \frac{1}{N} \sum_n \mathbb{E}[|F[n]|_F] &= \frac{1}{N} \sum_n \mathbb{E}[\text{tr}(F[n] F^*[n])] = N_t \\ R_x = \mathbb{E}[x x^*] &= \frac{E_s}{N_t} \mathbb{E}[F s s^* F^*] = \frac{E_s}{N_t} F[n] F^*[n] \end{aligned}$$

the singular value decomposition can be used to rewrite F :

$$F[n] = \underset{\substack{\uparrow \\ \text{Beam} \\ \text{Steering} \\ \text{Matrix}}}{U_F[n]} \underset{\substack{\uparrow \\ \text{Power} \\ \text{Allocation} \\ \text{Matrix}}}{\Sigma_F[n]} \underset{\substack{\uparrow \\ \text{Mixing} \\ \text{Matrix}}}{V_F^*[n]}$$

With:

- U_F, V_F unitary and respectively $N_t \times N_t, N_s \times N_s$
- $\Sigma_F = \begin{bmatrix} P^{1/2} \\ 0 \end{bmatrix}$ $N_t \times N_s$ matrix where $P^{1/2}$ is a square diagonal $N_s \times N_s$ matrix with the values $P_0, \dots, P_{N_s-1}$

3.5) Zero Forcing

Basic Case

$$w^*(z) \sqrt{G} h(z) = z^{-\Delta}$$

ideal equalizer:

$$w^*(z) = \frac{z^{-\Delta}}{\sqrt{G} h(z)} \rightarrow (h * w^*)[n] = \delta(n - \Delta) \xrightarrow{Z} h(z) w^*(z) = z^{-\Delta}$$

MIMO Case:

W_{ZF}^* is the **left inverse of H** that satisfies

$$W^*(z) \sqrt{G} H(z) = D(z)$$

Theorem 4 (Theorem on FIR left inverses).

A MIMO transfer function $H(z)$ has a FIR left inverse iff $H(z)$ has full column rank $\forall z \neq 0$

It also turns out that:

(!exam)

It turns out that an exact FIR ZF MIMO inverse exists for $N_r > N_t$ As long as the entries of H do not share a common root (coprime), then a FIR inverse exists

Rewrite:

$$W^*(z)\sqrt{G}H(z) = D(z) \xrightarrow{z^{-1}} \sqrt{G} \sum_{l=0}^{L_{eq}} W^*[l]H[n-l] = \text{diag}(\delta(n - \Delta_0), \dots, \delta(n - \Delta_{N_t-1}))$$

where:

- $D = \text{diag}(z^{-\Delta_0}, \dots, z^{-\Delta_{N_t-1}})$

$$\text{Look at single entry: } \sqrt{G} \sum_{l=0}^{L_{eq}} w_j^*[l]H[n-l] = d_j^*[n]$$

where:

- $d_j^*[n] = 0$ everywhere except j-th entry. Size $= N_r \times 1$
- $w_j^*[n] = W_{:,j}$ the j-th column of W . Size $N_t \times 1$

$$\text{stacked vectors to remove conv: } \sqrt{G}w_j^*T = d_j^*$$

where:

- $d_j^* = [d_j^*[0], \dots, d_j^*[L_{eq}]]^T$ of size $(L_{eq} + 1)N_r \times 1$
- $w_j^* = [w_j^*[0], \dots, w_j^*[L_{eq} + L]]^T$ of size $(L_{eq} + L + 1)N_t \times 1$
- T the classic toeplitz matrix but inverted of size $N_r(L_{eq} + 1) \times N_t(L + L_{eq} + 1)$

$$T = \begin{bmatrix} H[0] & \dots & H[L] & 0 & 0 & \dots & 0 \\ 0 & H[0] & \dots & H[L] & 0 & \dots & 0 \\ \vdots & & & \ddots & & & \vdots \\ 0 & \dots & \dots & 0 & H[0] & \dots & H[L] \end{bmatrix}$$

This system is solved using the Moore Penrose Pseudoinverse (MPP):

$$w_j = \frac{1}{\sqrt{G}}(T^*)^\dagger d_j$$

- **T is tall** then system underdetermined an ∞ solutions. Use MPP and see T^* is fat so $(T^*)^\dagger = T(T^*T)^{-1}$ and the solution is the minimum norm solution.

$$w_j = \frac{1}{\sqrt{G}}T(T^*T)^{-1}d_j$$

- **T is square** then there exists exactly one solution

$$w_j = \frac{1}{\sqrt{G}}(T^*)^{-1}d_j$$

- **T is fat** then the system is overdetermined and has no solutions. MPP finds the least squares solution with minimum norm. Since T^* is tall we have $(T^*)^\dagger = (TT^*)^{-1}T$ and the solution is

$$w_j = \frac{1}{\sqrt{G}}(TT^*)^{-1}Td_j$$

Same can be obtained with

$$W_{ZF} = \frac{1}{\sqrt{G}} (T^*)^\dagger D$$

Linear ZF eq in Frequency Flat Channel

Set $L_{eq} = L = 0$ then $T = H$ and H is tall. By MPP

$$W_{ZF} = \frac{1}{\sqrt{G}} H(H^*H)^{-1} D \rightarrow W_{ZF}^* = \frac{1}{\sqrt{G}} (H^*H)^{-1} H^*$$

and thus the equalizer has no error:

$$W_{ZF}^* \sqrt{G} H = \frac{1}{\sqrt{G}} (H^*H)^{-1} H^* \sqrt{G} H = (H^*H)^{-1} H^* H = I$$

LMMSE IID

$$W^{MMSE} = \frac{1}{\sqrt{G}} \left(TT^* + \frac{N_t}{SNR} I \right)^{-1} TD$$

$$E = \frac{E_s}{N_t} I - \frac{E_s}{N_t} D^* T^* \left(TT^* + \frac{N_t}{SNR} I \right)^{-1} TD$$

(!exam) if there is no noise ($SNR \rightarrow \infty$) then the error goes to 0:

$$E \xrightarrow{SNR \rightarrow \infty} \frac{E_s}{N_t} I - \frac{E_s}{N_t} D^* T^* (TT^* + 0) TD = \frac{E_s}{N_t} I - \frac{E_s}{N_t} D^* T^* \underbrace{(TT^*)}_{I} TD = \frac{E_s}{N_t} I - \frac{E_s}{N_t} I = 0$$

3.6) MIMO

Transmitter

Coding happens in frequency domain, transmitter sends block of K with L prefix symbols.

$$x_f[0], \dots, x_f[K-1] \rightarrow \bar{x}_t[-L], \dots, \bar{x}_t[K-1] \text{ where } \bar{x}_t[n] = \begin{cases} x_t[n+K] & n = -L, \dots, -1 \\ x_t[n] & n = 0, \dots, K-1 \end{cases}$$

Verify (!exam):

$$\begin{aligned} \bar{x}_t[n] &= \frac{1}{K} \sum_{k=0}^{K-1} x_f[k] e^{j\frac{2\pi}{K}nk} \\ &= \frac{1}{K} \sum_{k=0}^{K-1} x_f[k] e^{j\frac{2\pi}{K}nk} \underbrace{e^{j\frac{2\pi}{K}Kk}}_{=1} \\ &= \frac{1}{K} \sum_{k=0}^{K-1} x_f[k] e^{j\frac{2\pi}{K}(K+n)k} = x_t[K+n] \end{aligned}$$

This is just the **periodicity property of the DFT**.

□

Channel:

IDFT of cyclic prefix needs previous block of symbols to be computed, so **cyclic prefix removal** is done.

$$y_t[n] = \begin{cases} \sqrt{G} \sum_{l=0}^L h_t[l] \bar{x}_t[n-l] & n = 0, \dots, K-1 \\ \sqrt{G} \sum_{l=0}^{L+n} h_t[l] \bar{x}_t[n-l] + \sqrt{G} \sum_{l=L+n+1}^L h_t[l] \underbrace{\bar{x}_t[n-l]}_{\text{Out of Block}} & n = -L, \dots, -1 \end{cases}$$

Proof out of block:

Rewrite sum with $m = l - (L + n + 1)$ so the summation becomes

$$\begin{aligned}
 \sum_{l=L+n+1}^L h_t[l] \bar{x}_t[n-l] &= \sum_{m=0}^{-n-1} h_t[L+n+1+m] \bar{x}_t[n-(L+n+1+m)] \\
 &= \sum_{m=0}^{-n-1} h_t[L+n+1+m] \bar{x}_t[-L-1-m] \\
 &= \sum_{m=0}^{-n-1} h_t[L+n+1+m] \bar{x}_{t-1}[K-L-1-m] \\
 &= \sum_{l=L+n+1}^L h_t[l] \bar{x}_{t-1}[n-l+K]
 \end{aligned}$$

where x_{t-1} is the previous block.

□

Receiver:

The receiver computes the DFT using the cyclic prefix, which is a circular convolution

$$y_t[n] = \sqrt{G} \sum_{l=0}^{L-1} \bar{h}_t[n] \bar{x}_t[n-l] = \sqrt{G} \sum_{l=0}^{L-1} \bar{h}_t[n] x_t[(n-l)_K] \xrightarrow{\mathcal{F}} x_f[k] = \sqrt{G} h_f[k] x_f[k]$$

Proof of circular convolution is product in frequency (!exam):

$$\begin{aligned}
 \tilde{y}_f[n] &= \sum_{p=0}^{K-1} y_t[p] e^{-j \frac{2\pi}{K} np} \\
 &= \sum_{p=0}^{K-1} \sqrt{G} \sum_{l=0}^L h_t[l] \bar{x}_t[p-l] e^{-j \frac{2\pi}{K} np} \\
 &= \sum_{p=0}^{K-1} \sqrt{G} \sum_{l=0}^L h_t[l] \frac{1}{K} \sum_{k=0}^{K-1} x_f[k] e^{j \frac{2\pi}{K} (p-l)k} e^{-j \frac{2\pi}{K} np} \\
 &= \sqrt{G} \sum_{k=0}^{K-1} \left[\left(\sum_{l=0}^{K-1} h_t[l] e^{-j \frac{2\pi}{K} lk} \right) x_f[k] \left(\frac{1}{K} \sum_{p=0}^{K-1} e^{j \frac{2\pi}{K} p(k-n)} \right) \right] \\
 &= \sqrt{G} \sum_{k=0}^{K-1} h_f[k] x_f[k] \delta(k-n) \\
 &= \sqrt{G} h_f[n] x_f[n]
 \end{aligned}$$

□

OFDM Estimation:

With absence of noise a simple eq is used:

$$\hat{x}_f[k] = \frac{\tilde{y}_f[k]}{\sqrt{G} h_f[k]}$$

With noise notice that $v_t[n] \approx v_f[k]$ (TODO Appendix C1.6) with the same ZF eq the variance of the output noise becomes

$$\frac{N_0}{G} \frac{1}{|h_f[k]|^2}$$

3.7) Channel Estimation

For channel estimation some well known signals, called pilot signals $p[0], \dots, p[N_p - 1]$ where $p[\cdot]$ is a $N_t \times 1$ vector. These signals are sent through a L tap channel $\sqrt{G}H[l]$ and from the received signals a estimation on the channel can be made.

$$P = \begin{bmatrix} p[L] & p[L+1] & \dots & p[N_p-1] \\ p[L-1] & p[L+2] & \dots & p[N_p-2] \\ \vdots & \vdots & \vdots & \\ p[0] & p[1] & \dots & p[N_p-L-1] \end{bmatrix}$$

$$Y = \sqrt{G} H P + V \rightarrow \text{vec } Y = \sqrt{G} P_{\otimes} \text{vec } H + \text{vec } V$$

Proof:

$$\begin{aligned} \text{vec } Y &= \sqrt{G} \text{vec } (HP) + \text{vec } V \\ &= \sqrt{G} \text{vec } (HIP) + \text{vec } V \\ &= \sqrt{G} \underbrace{(P^T \otimes I)}_{P_{\otimes}} \text{vec } H + \text{vec } V \end{aligned}$$

□

ML and Least Squares Estimation & LMMSE

In AWGN noise ML and LS estimators are equivalent

Proof:

$$\text{vec}[\hat{H}](y) = \arg \max_h f_{y|h}(y|h) = \arg \max_h \frac{1}{\det(\pi R)} e^{-\frac{|y-\mu|^2}{R}} \rightarrow \arg \min_h |y - \mu|^2$$

since constants can be removed from max and the exponent is monotonically decreasing

□

Notice that $y|h \sim \mathcal{N}_{\mathbb{C}}(\sqrt{G}P_{\otimes}h, N_0)$

	ML/Least Squares	LMMSE
Estimator	$\text{vec}[\hat{H}](y) = \frac{1}{\sqrt{G}} (P_{\otimes}^* P_{\otimes})^{-1} P_{\otimes}^* y$ $\hat{H} = \frac{1}{\sqrt{G}} Y P_{\otimes}^* (P_{\otimes}^* P_{\otimes})^{-1}$	$\text{vec} \hat{H} = \left((\sqrt{G} P_{\otimes} (N_0 R_{\text{vec} \hat{H}}^{-1} + G P_{\otimes}^* P_{\otimes})^{-1})^* y \right)$
MMSE	$E = \frac{N_0}{G} (P_{\otimes}^* P_{\otimes})^{-1}$	$E = R_{\text{vec} \hat{H}} - G R_{\text{vec} \hat{H}}^* P_{\otimes}^* (G P_{\otimes} R_{\text{vec} \hat{H}} P_{\otimes}^* + N_0 I)^{-1} P_{\otimes} R_{\text{vec} \hat{H}}$

(!exam)

Additionally measure $h_f[k]$ by estimating in frequency with LS or LMMSE, then do IDFT and the first L samples are the L taps, the remaining can be used to estimate noise. When you truncate $\hat{h}_t[n]$ at L and do DFT of $h_t[n]$ and get a better estimate of h in frequency.

3.8) Additional Exercises

Example 2.15, 2.16 (Two Tap Channel ZF Equalizer).

For a two tap channel where $\sqrt{G}h[0] = 1, \sqrt{G}h[1] = -a$ find an IIR equalizer with $\Delta = 1$.

First find transform of $h[n] = \delta(n) - a\delta(n-1) \xrightarrow{z} 1 - az^{-1}$
therefore we have:

$$w(z) = \frac{1}{\sqrt{G}} \frac{z^{-1}}{1 - az^{-1}} \xrightarrow{z^{-1}} w[l] = \frac{1}{\sqrt{G}} a^{l-1} u[l-1]$$

Where an appropriate FIR equalizer can be found by truncating.

Example 2.22, 2.23 (Simple OFDM Exercises).

Let $T_{\text{OFDM}} = 3.2\mu\text{s}$ with $K = 256 = 2^8$ subcarriers and a cyclic prefix of length $L = 64 = 2^6$. Find the chip period T , the passband bandwidth, the subcarrier spacing, and the guard interval of the cyclic prefix.

Recall $T_{\text{OFDM}} = (K + L)T \rightarrow T = 10\text{ns} \rightarrow B = 1/T = 100\text{MHz}$

The spacing is $1/KT = 390.6\text{KHz}$ and the guard interval is $LT = 0.64\mu\text{s}$. **Impulse response of this duration or less can be handled.**

Now let the subcarrier spacing be 15kHz and every 5MHz of bandwidth accomodates $K = 300$ subcarriers. What is the spectral efficiency penalty?

$$P_{\min} = \frac{B_{\text{final}}}{B_{\text{tot}}} = \frac{K \cdot 15 \cdot 10^3}{5 \cdot 10^6} = 0.9$$

The real penalty also includes the guard interval so

$$P = P_{\min} \frac{K}{K + L}$$

Example 2.24 (Least Squares Estimatiior For a Frequency Flat SISO Channel).

For a SISO system with frequency flat ($L = 0$) channel estimate the channel and the MMSE with a single pilot $p = e^{j\phi}$.

The single pilot, SISO and frequency flat case transforms the problem into a simple 1D case. In fact $P = p$ and $H = h$ from here also $P_{\otimes} = p$

Recall the solution of the least squares problem:

$$\text{vec}[\hat{H}] = \frac{1}{\sqrt{G}} (P_{\otimes}^* P_{\otimes})^{-1} P_{\otimes}^* y$$

First calculate $P_{\otimes}^* P_{\otimes} = E_s$, then the full matrix multiplications become $\frac{\sqrt{E_s}}{E_s} e^{-j\phi} = \frac{e^{-j\phi}}{\sqrt{E_s}}$.

Then

$$\hat{h} = \frac{e^{-j\phi}}{\sqrt{GE_s}} y$$

Same for the MMSE:

$$E = \frac{N_0}{G} (P_{\otimes}^* P_{\otimes})^{-1} \rightarrow \text{MMSE} = \frac{N_0}{GE_s} = \frac{1}{\text{SNR}}$$

Also in the case $N_p > 1$ the MMSE becomes

$$MMSE = \frac{1}{N_p SNR}$$

Example 2.29 (LMMSE Estimator For a Frequency Flat SISO Channel).

As before $L = 0$, SISO and with one pilot, then recall

$$W_{MMSE} = \sqrt{G}P_{\otimes}(N_0R_{\text{vec}\hat{H}}^{-1} + GP_{\otimes}^*P_{\otimes})^{-1}$$

Which in the 1d case becomes:

$$W_{MMSE} = \frac{\sqrt{GE_s}}{N_0 + GE_s} e^{j\phi} = \frac{\sqrt{GE_s}/N_0}{1 + GE_s/N_0} \frac{\sqrt{GE_s}}{\sqrt{GE_s}} e^{j\phi} = \frac{SNR}{1 + SNR} \frac{e^{j\phi}}{\sqrt{GE_s}}$$

then $\text{vec}\hat{H} = W_{MMSE}^* y$ means that

$$\hat{h} = \frac{SNR}{1 + SNR} \frac{e^{j\phi}}{\sqrt{GE_s}} y$$

The MMSE then becomes, $E = R_{\text{vec}\hat{H}} - GR_{\text{vec}\hat{H}}^* P_{\otimes}^* (GP_{\otimes} R_{\text{vec}\hat{H}} P_{\otimes}^* + N_0 I)^{-1} P_{\otimes} R_{\text{vec}\hat{H}}$ and in 1d (works for $N_p \geq 1$)

$$MMSE = 1 - GE_s(GE_s + N_0)^{-1} = \frac{1}{1 + N_p SNR}$$

Which is *always smaller than the least squares case*

LMMSE ESTIMATION COINCIDES WITH LEAST SQUARES IN IID MIMO

if $P_{\otimes}$ Is unitary, then $P_{\otimes} P_{\otimes}^* = \frac{E_s}{N_t} I$ and thus

$$\begin{aligned} E &= I - GP_{\otimes}^* (GP_{\otimes} P_{\otimes}^*)^{-1} P_{\otimes} = I - GP_{\otimes}^* \left(\frac{GE_s}{N_t} + N_0 \right)^{-1} I P_{\otimes} \\ &= I - GP_{\otimes}^* P_{\otimes} \left(\frac{GE_s}{N_t} \left(1 + \frac{N_t}{SNR} \right) \right)^{-1} \\ &= I - \frac{GE_s}{N_t} \frac{1}{\frac{GE_s}{N_t} \left(1 + \frac{N_t}{SNR} \right)} I \\ &= \frac{1}{1 + \frac{SNR}{N_t}} I \end{aligned}$$

4) Chapter 3: Channel Modeling

Large Scale:

Link Budget

$$\text{Log Scale: } P_r[dB] = P_t[dB] + \underbrace{G_t[dB] + G_r[dB] - (L_p[dB] + \chi[dB])}_{\text{large scale gain } G}$$

$$\text{Linear Scale: } P_r = P_t \cdot \frac{G_t G_r}{L_p \chi}$$

Path Loss

$$L_P(D) = K_{ref} \left(\frac{D}{D_{ref}} \right)^\eta \rightarrow L_p(D)[dB] = K_{ref}[dB] + 10\eta \log_{10} \left(\frac{D}{D_{ref}} \right)$$

Free Space Path Loss

$$L_P(D)[dB] = 20 \log_{10} \left(\frac{4\pi D}{\lambda_c} \right)$$

Small Scale:

With sinusoid signals:

$$c(\tau) = \sum_q A_q e^{-j2\pi f_c \tau_q} \delta(\tau)$$

Proof:

If $\max_q \tau_q \gg 1/B$ the delays act as a phase shift:

$$x(t) = e^{j2\pi f_c t} \xrightarrow{\text{channel}} y(t) = \sum_q A_q e^{j2\pi f_c (t - \tau_q)} = \sum_q A_q e^{-j2\pi f_c \tau_q} x(t)$$

□

Model the small-scale channel response as a locally stationary complex rv $h = |h|e^{j\angle h} \sim \mathcal{N}_{\mathbb{C}}(0, 1)$, then the channel has these 3 properties:

- Uniform Phase: $\phi = \angle h \sim U(0, 2\pi)$
- Rayleigh Magnitude: $f_{|h|}(\xi) = \xi e^{-\frac{1}{2}\xi^2}$
- Exponential magnitude squared: $f_{|h|^2}(\xi) = e^{-\xi}$

If the multi path components dominate over the LoS, then rice fading can be used:

$$f_{|h|}(\xi) = 2(K+1)\xi e^{-(K+1)\xi^2} I_0 \left(2\sqrt{K(K+1)}\xi \right)$$

where

- $K = 0$: rice equals Rayleigh
- $K \rightarrow \infty$: rice is deterministic

Space Selectivity (Power Angle Spectrum):

The PAS works with the **azimuth** angle and describes the Rx power as a function of the angle.

By integrating the PAS in the angle the total received power is obtained.

Space Correlation

Again, supposing x a complex exponential being received at a given location

$$y(t) = \sqrt{G} \int_{2\pi} \sqrt{\mathcal{P}(\theta)G_r(\theta)} x(t - D(\theta)/C) d\theta$$

$$y = \sqrt{G} \int_{2\pi} \sqrt{\mathcal{P}(\theta)G_r(\theta)} x e^{-j\phi(\theta)} d\theta$$

Where $\phi(\theta) = 2\pi D(\theta)/\lambda_c$

At a second location distanced Δ_d from the first one we have that $\phi'(\theta) = \phi(\theta) + 2\pi\Delta_d \cos(\theta)/\lambda_c$

Time Correlation

Let the **maximum doppler shift** be

$$\nu_M = \frac{v}{c} f_c$$

and the **doppler shift**

$$\nu = \nu_M \cos(\theta)$$

The previous result can be seen as a signal sent while moving, so by integrating over the max doppler shift we end up with the fourier transform

	General	Clarke-Jakes
Space Selectivity: PAS	$\mathcal{P}(\theta)$	$\mathcal{P}(\theta) = \frac{1}{2\pi}$
Space Correlation: phase $\phi(\theta) \sim U[-\pi, \pi]$	$R_h(\Delta_d) = \mathbb{E}[h_0 h_1^*] = \int_{-\pi}^{\pi} \mathcal{P} G e^{j2\pi\Delta_d \cos(\theta)\lambda_c} d\theta$	$R_h(\Delta_d) = J_0(2\pi\Delta_d/\lambda_c)$ $D_c = 0.38\lambda_c$
Time Correlation: Time and Frequency	$R_h(\Delta_t = \Delta_d/v)$ $S_h(\nu) = \frac{\mathcal{P}(-\arccos(\nu/\nu_M)) + \mathcal{P}(\arccos(\nu/\nu_M))}{\sqrt{\nu_M^2 - \nu^2}}, \nu \in [-\nu_m, \nu_m]$	$R_h(\Delta_t) = J_0(2\pi\nu_M\Delta_t)$ $S_h(\nu) = \frac{1}{\pi\sqrt{\nu_M^2 - \nu^2}}$

Discrete Time Model

In this case

$$R_h(lT) = \frac{1}{T} S_h\left(\frac{\nu}{T}\right)$$

In the block fading model the number of single carrier symbols per block is

$$N_c = \frac{T_c}{T} = BT_c \xrightarrow{\text{IID Blocks}} \frac{1}{2\nu_m T}$$

Moreover in OFDM we have that

$$N_c = \frac{K}{K+L} T_c B_c \approx T_c B_c$$

Coherence Time

The coherence time is the measure of the time duration over which the wireless channel is considered "static" or constant.

$$T_c \gg T_s \quad T_c = \frac{1}{2\nu_M} \propto \frac{1}{\nu_M}, \quad \nu_m \ll B$$

Frequency Selective

$$\max_{i,j} |\tau_i(t) - \tau_j(t)| = \tau_m \stackrel{\Delta}{=} \text{Delay Spread}$$

$$\boxed{\text{Frequency Flat iff: } \tau_m \ll T_s \iff T_d \ll T_s \iff B_c \gg B} \\ \implies \tau_i(t) \approx \tau_j(t) \approx \hat{\tau} \quad \forall i, j \rightarrow h(\tau) = h\delta(\tau)$$

Here **gain is constant across frequency and the estimation of $g(t)$ suffices to compensate for phase and amplitude displacements**

The **received power as a function of delay (normalized)** is called **Power Delay profile (PDP)** and can be interpreted as a PDF.

$$S_h(\tau) = \mathbb{E}[|h(\tau)|^2]$$

It has the following mean and variance:

$$\mu_\tau = \int \tau S_h(\tau) d\tau \quad \longrightarrow \sum_{q=0}^{Q-1} \tau_q P_q$$

$$\sigma_\tau^2 = T_d = \sqrt{\int (\tau - \mu_\tau)^2 S_h(\tau) d\tau} \quad \longrightarrow \sqrt{\sum_{q=0}^{Q-1} (\tau_q - \mu_q)^2 P_q}$$

If the symbols are uncorrelated, then the fourier transform of the PDP is the frequency domain correlation $R_h(\Delta_f)$ (depends only on Δ_f , wide sense stationary)

$$R_h(\Delta_f) = \int S_h(\tau) e^{j2\pi\Delta_f\tau} d\tau$$

if $\mathbb{E}[h(\tau_0)h(\tau_1)] = 0$ uncorrelated scattering
And from here we have that

$$B_c \propto \frac{1}{T_d}$$

Moreover wireless fading is **underspread**:

$$T_d \ll T_c, \quad \nu_M \ll B_c \rightarrow B_c T_c \gg 1$$

4.2) Additional Exercises

Example 3.13 (Clarke-Jakes PAS Coherence Time and Distance).

Let $f_c = 2 \text{ GHz}$ and a velocity $v = 5 \text{ km/h}$, we have:

$$\lambda_c = \frac{c}{f_c} = 15 \text{ cm} \rightarrow D_c = 5.7 \text{ cm}$$

And the coherence time is (recall $m/s = 3.6 \cdot km/h$)

$$T_c = \frac{1}{2\nu_m} = \frac{1}{2 \frac{v}{c} f_c} = 41 \text{ ms} = \frac{D_c}{0.76 \cdot v} \approx \frac{D_c}{v}$$

Example 3.23 (Continuous PDP distribution Calculation).

Let the PDP be

$$S_h(\tau) = K_e \exp \{-4 \cdot 10^5 \tau\}, \tau \in [0, 10] \mu s$$

First set K_e so to normalize the PDP:

$$1 = \int_0^{10 \cdot 10^{-6}} S_h(\tau) d\tau \approx 2.45 \cdot 10^{-6} K_e \rightarrow K_e = 4.08 \cdot 10^5$$

The mean is:

$$\begin{aligned} \mu_\tau &= \int_0^{10} \tau K_e \exp \{-4 \cdot 10^5 \tau\} d\tau = K_e \left[-\frac{e^{-4 \cdot 10^5 \tau}}{4 \cdot 10^5} \left(\tau + \frac{1}{4 \cdot 10^5} \right) \right] \Big|_0^{10} \\ &\approx 5.68 \cdot 10^{-12} K_e = 2.31 \mu s \end{aligned}$$

And the RMS delay spread can be calculated in closed form but just the result is shown:

$$T_d \approx 2.08 \mu s$$

This means that a symbol time $T \gg 2.08 \mu s$ is required for a frequency flat approximation.

$$B_c \approx \frac{1}{T_d} = 480 kHz$$

and therefore supposing we want at least 1000 symbols as the underspread at 2 GHz, it is possible to go as fast as

$$B_c T_c = \frac{B_c}{2\nu_M} = 1000 \rightarrow \nu_m = \frac{f_c v}{c} = \frac{B_c}{2000} \rightarrow v = \frac{c B_c}{2000 f_c} = 36 m/s = 130 km/h$$